

**Multiple Choice:**

1. Which of the following is not an example of external noise?
 - A. Fluorescent light
 - B. Solar emission
 - C. Resistor noise
 - D. Lightning
2. Why does a 5-kHz square wave require a greater bandwidth than a 2-kHz sinewave?
 - A. The square wave has a larger frequency than the sinewave.
 - B. The square wave has an infinite number of harmonics, whereas the sinewave has only one.
 - C. The square wave is much easier to cover by noise.
 - D. The square wave must also include an offset (dc) voltage, whereas the sinewave does not.
3. The collection of sinusoidal frequencies present in a modulated carrier is called its:
 - A. frequency-domain representation
 - B. spectrum
 - C. Fourier series
 - D. All of the above
4. One of the following is not useful quantity for comparing the noise performance of the receiver
 - A. Input noise voltage
 - B. Equivalent noise resistance
 - C. Noise temperature
 - D. Noise figure
5. So called " $1/f$ " noise is also called:
 - A. random noise
 - B. white noise
 - C. pink noise
 - D. partition noise
6. A system is said to be shift invariant only if _____.
 - A. a shift in the input signal also results in the corresponding shift in the output
 - B. a shift in the input signal does not exhibit the corresponding shift in the output
 - C. a shifting level does not vary in an input as well as output
 - D. a shifting at input does not affect the output
7. Indicate the noise whose source is in a category different from that of the other three.
 - A. solar noise
 - B. cosmic noise
 - C. atmospheric noise
 - D. galactic noise
8. Determine the shot noise for a diode with a forward bias of 1.40 mA over an 80 kHz bandwidth. ($q = 1.6 \times 10^{-19}$ raised to minus 19 C)
 - A. 0.006 μ A
 - B. 0.0006 μ A
 - C. 0.06 μ A
 - D. 0.6 μ A
9. Damped sinusoids are _____.
 - A. sinusoid signals multiplied by growing exponentials
 - B. sinusoid signals divided by growing exponentials
 - C. sinusoid signals multiplied by decaying exponentials
 - D. sinusoid signals divided by decaying exponentials
10. Noise in a communication system originates in:
 - A. the sender
 - B. the channel
 - C. the receiver
 - D. All of the above
11. According to Rayleigh's theorem, it becomes possible to determine the energy of a signal by _____.
 - A. estimating the area under the square root of its amplitude spectrum
 - B. estimating the area under the square of its amplitude spectrum
 - C. estimating the area under the one-fourth power of its amplitude spectrum
 - D. estimating the area exactly half as that of its amplitude spectrum
12. Signal-to-noise ratio is calculated as:
 - A. signal voltage divided by noise voltage
 - B. signal power divided by noise power
 - C. first add the signal power to the noise power, then divide by noise power
 - D. None of the above
13. Which among the below mentioned transform pairs is/are formed between the auto-correlation function and the energy spectral density, in accordance to the property of Energy Spectral Density (ESD)?
 - A. Laplace Transform
 - B. Z-Transform
 - C. Fourier Transform
 - D. All of the above
14. Indicate the false statement. The square of the thermal noise voltage generated by a resistor is proportional to
 - A. its resistance
 - B. its temperature
 - C. Boltzmann's constant
 - D. the bandwidth over which it is measured
15. A system is said to be shift invariant only if _____.
 - A. a shift in the input signal also results in the corresponding shift in the output
 - B. a shift in the input signal does not exhibit the corresponding shift in the output
 - C. a shifting level does not vary in an input as well as output
 - D. a shifting at input does not affect the output
16. "Pink" noise has:
 - A. equal power per Hertz
 - B. constant power
 - C. equal power per octave
 - D. None of the above



17. Which among the below mentioned transform pairs is/are formed between the auto-correlation function and the energy spectral density, in accordance to the property of Energy Spectral Density (ESD)?

- A. Laplace Transform
- B. Z-Transform
- C. Fourier Transform
- D. All of the above

18. The power density of "flicker" noise is:

- A. the same at all frequencies
- B. greater at low frequencies
- C. greater at high frequencies
- D. the same as "white" noise

19. Which theorem states that the total average power of a periodic signal is equal to the sum of average powers of the individual Fourier coefficients?

- A. Parseval's Theorem
- B. Rayleigh's Theorem
- C. Both A and B
- D. None of the above

20. In a communication system, noise is most likely to affect the signal

- A. at the transmitter
- B. in the channel
- C. in the information source
- D. at the destination

21. What would happen if the value of term $[(m-x) / (\sigma \sqrt{2})]$ increases in the expression of Gaussian CDF?

- A. Complementary error function also goes on increasing
- B. Complementary error function goes on decreasing
- C. Complementary error function remains constant or unchanged
- D. Cannot predict

22. SINAD is calculated as:

- A. signal voltage divided by noise voltage
- B. signal power divided by noise power

- C. first add the signal power to the noise power, then divide by noise power
- D. None of the above

23. Shot noise is generated in:

- A. transistors and diodes
- B. copper wire
- C. resistors
- D. None of the above

24. A diode generator is required to produce 12 microvolts of noise in a receiver with an input impedance of 75 ohms and a noise power bandwidth of 200 kHz. Determine the current through the diode in milliamperes.

- A. 390 mA
- B. 397 mA
- C. 396 mA
- D. 398 mA

25. Noise Figure is a measure of:

- A. how much noise is in a communications system
- B. how much noise is in the channel
- C. how much noise an amplifier adds to a signal
- D. signal-to-noise ratio in dB

END